Proxmark Indala → T5577 Clone Playbook

This is a concise, copy-paste-ready checklist for cloning an Indala 125kHz tag to a blank T5577 using a Proxmark3 client. Run every command from the **Proxmark3 client** (you'll be at the `pm3>` prompt). Keep the USB connected during reads/writes.

Quick explanation (why you may see mixed detections)

- `lf search` runs multiple decoders. It may *display* an Indala decode (payload/ID interpretation) while the low-level *chip* detection looks like `EM4x05 / EM4x69` based on timing or header patterns.
- A T5577 (or EM4305) clone **emulates** the higher-level Indala payload, but the proxmark may still detect the underlying modulation/chip pattern differently. "Odd size" / "false positive" messages are common with cloned tags and cheap blanks.

Pre-checks (physically)

1. Use a data-capable USB cable to PC. No hubs preferred. Keep battery out if removable.
2. Remove other tags from around the antenna.
3. Place the source tag or blank flat and centered on the coil when prompted.

Canonical workflow (run one command at a time)

0 — Basic device check (PowerShell / terminal where proxmark runs)

```
pm3> hw version
```

1 — Tune / quick detect

```
pm3> hw tune
pm3> lf search
```

If `lf search` shows `Valid Indala ID found!`, note the RAW value (e.g. `a0000000e3f00d51`).

2 — Clone using raw (fast, non-interactive)

Physical: Place **blank T5577** on coil before running write command.

```
pm3> lf indala clone -r a0000000e3f00d51
```

This will map to T5577 blocks and attempt the write. Check for a `Done!` message.

3 — Verify tag mapping & chip

With the T5577 on coil:

```
pm3> lf t55xx detect
pm3> lf t55xx read --blk 1 --r1
pm3> lf t55xx read --blk 2 --r1
pm3> lf search
```

- Use `--r1` because most Indala clones use the long leading reference downlink. If `detect` fails, run `lf t55xx config` (see below).

4 — If readback is wrong: configure and re-write (safe sequence)

```
pm3> lf t55xx config --PSK3 --rate 16 --r1
pm3> lf t55xx write --blk 0 --data 00088040
```

```
pm3> lf t55xx write --blk 1 --data A0000000
pm3> lf t55xx write --blk 2 --data E3F00D51
pm3> lf t55xx read --blk 0 --r1
pm3> lf t55xx read --blk 1 --r1
pm3> lf t55xx read --blk 2 --r1
pm3> lf search
```

- If the blank uses an alternate page mapping try `--pg1` on `read`/`write` and re-check:

```
pm3> lf t55xx write --blk 1 --data A0000000 --pg1
pm3> lf t55xx read --blk 1 --r1 --pg1
```

5 — Offset handling (if data seems "shifted")

- Set offset globally (example uses detected offset 53):

```
pm3> lf t55xx config --offset 53
pm3> lf t55xx read --blk 1 --r1
```

- Some firmwares do not accept `-o` on read; use `config` to set offset.

6 — Low-level manual writes (if needed)

If the interactive clone fails, force these block writes (run only on tags you own):

```
pm3> lf t55xx write --blk 0 --data 00088040
pm3> lf t55xx write --blk 1 --data A0000000
pm3> lf t55xx write --blk 2 --data E3F00D51
```

Then read back with `lf t55xx read --blk X --r1`.

Troubleshooting checklist (ordered)

1. `hw tune` — poor coupling = failed writes. Reposition tag and re-tune.
2. Re-run `lf t55xx detect` — if it fails, set `lf t55xx config --PSK3 --rate 16 --r1`.
3. If readback shows garbage: try `--pg1` and different offsets (via `lf t55xx config --offset N`).
4. If writes appear to succeed but readback never matches, try a different blank (some cheap T5577s are flaky).
5. If proxmark shows Indala but reader rejects clone: adjust target type (`--em` for EM4305) or try a different blank vendor.

Safe commands summary (copy-paste)

All commands in one place for quick copy-paste at `pm3>`:

```
hw version
hw tune
lf search

# Read & save source (optional)
lf indala read -f indala_raw.bin

# Fast clone (use raw from lf search)
lf indala clone -r <raw_hex>

# Configure T55xx
lf t55xx config --PSK3 --rate 16 --r1
lf t55xx detect

# Force writes
lf t55xx write --blk 0 --data 00088040
lf t55xx write --blk 1 --data A0000000
lf t55xx write --blk 2 --data E3F00D51

# Readback
```

```
lf t55xx read --blk 0 --r1
lf t55xx read --blk 1 --r1
lf t55xx read --blk 2 --r1

# Page1 variants
lf t55xx write --blk 1 --data A0000000 --pg1
lf t55xx read --blk 1 --r1 --pg1

# Extra helpers
lf t55xx read --help
lf t55xx write --help
lf t55xx config --help
lf indala clone --help
```

Field test

After the proxmark verifies the clone, **test at the real Indala reader**. If it opens the door, you're done — proxmark readback quirks don't matter.

Notes & ethics

- Clone only tags you own or have permission to clone. Don't attempt to bypass access controls without explicit authorization.
- If you need higher reliability, buy better blanks (T5577 from a reputable RFID supplier or EM4305 blanks).

If you want, I can also export this as a downloadable file (PDF or .md) and add a short hardware checklist (battery, cable, spare blanks) — say which format you prefer.

Why the tag sometimes shows "Indala" but `lf search` / chipset detection shows EM4x05 (or similar):

- `lf search` runs many decoders and will show any *plausible* interpretation of the captured signal. It can therefore display an **Indala** decode (the human-friendly ID) even when the underlying *chip/modulation* looks like an EM4x05/EM4x69.
- A T5577 (or EM4305) clone **emulates the Indala payload**, but the proxmark's chipset-detection may pick up timing/header patterns and label the physical side as EM4x05 — that's not an error, it's two different layers of interpretation (payload vs chipset).
- Cheap blanks add noise: they often require different offsets or page mappings, so your Proxmark's decoders will disagree and print "odd size / false positive." The important test is the real reader: if it accepts the clone, you're done, even if proxmark's dumps look messy.